

CODTECH IT SOLUTIONS PRIVATE LIMITED

Data Analytics Internship — Project Report

MINI PROJECT REPORT

Data Analytics

TOPIC: CUSTOMER FEEDBACK SENTIMENT

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Organization	Codtech IT Solutions Private Limited
Internship Period	03 June 2026 – 15 July 2026
Duration	6 Weeks
Domain	Data Analytics
Dataset Used	Women's Clothing E-Commerce Reviews.csv
Tools / Language	Python, Pandas, NLTK, Scikit-learn, Matplotlib, Seaborn, WordCloud

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1. Problem Statement and Background

Customer reviews are one of the richest sources of feedback available to an e-commerce business, but the volume and unstructured nature of free-text reviews make manual analysis impractical at scale. Understanding what drives a customer to recommend a product — and predicting recommendation status directly from review text — allows a business to monitor customer experience, flag weak product categories, and prioritise catalogue decisions without waiting on slow manual review audits.

This project analyses the Women's Clothing E-Commerce Reviews dataset with two primary objectives:

- **Review Analytics:** To explore customer ratings, sentiment, department- and class-level performance, and the language customers use in positive versus negative reviews.
- **Recommendation Prediction:** To build and compare machine learning and NLP models that predict whether a customer recommends a product (Recommended IND = 1) directly from the cleaned review text, enabling automated monitoring of customer sentiment at scale.

By identifying the departments, classes, and language patterns associated with low recommendation rates, this analysis aims to help the business target weak product lines, understand common complaint themes, and prioritise customer-experience improvements.

2. Dataset Description

The Women's Clothing E-Commerce Reviews dataset is a widely-used retail text-analytics dataset containing real customer reviews and structured product/customer attributes. Key characteristics:

Filename	Women's Clothing E-Commerce Reviews.csv
Total Rows (Reviews)	23,486
Total Columns	11 (before feature engineering)
Key Columns	Clothing ID, Age, Title, Review Text, Rating, Recommended IND, Positive Feedback Count, Division Name, Department Name, Class Name
Target Variable	Recommended IND (binary: 1 = Recommended, 0 = Not Recommended)
Task Type	Binary Text Classification (NLP) + Exploratory Review Analytics

Key Column Descriptions

- **Title / Review Text:** Free-text fields combined into a single Full Review field and cleaned for NLP modelling.
- **Rating:** 1–5 star rating given by the customer; used to derive a three-class Sentiment label.
- **Recommended IND:** Binary flag — the primary target variable for classification.
- **Positive Feedback Count:** Number of other customers who found the review helpful.
- **Division Name / Department Name / Class Name:** Hierarchical product categorisation used for group-level performance comparisons.
- **Age:** Customer age, used to derive an Age Group feature for demographic analysis.

3. Methodology

3.1 Data Acquisition and Loading

The dataset was loaded from Google Drive into a Pandas DataFrame using `pd.read_csv()`. Initial inspection (`df.shape`, `df.head()`, `df.info()`) confirmed a structure of 23,486 rows across 11 original columns.

3.2 Data Cleaning

- Duplicate rows were checked using `df.duplicated().sum()` and removed with `drop_duplicates()`.
- The redundant index column (Unnamed: 0) was dropped.
- Missing Title and Review Text values were filled with empty strings so text processing could proceed without errors.
- Missing categorical values (Division Name, Department Name, Class Name) were filled with the label "Unknown".
- Missing numeric values (Clothing ID, Age, Rating, Recommended IND, Positive Feedback Count) were imputed using the column median.
- Title and Review Text were concatenated into a single Full Review field for downstream NLP processing.

3.3 Text Preprocessing (NLP)

A dedicated text-cleaning pipeline was built with an NLTK-based path and a scikit-learn stop-word fallback:

- Text was lower-cased and stripped of punctuation and digits.
- Tokenisation was performed with NLTK's `word_tokenize` (with a regex fallback if NLTK resources were unavailable).
- Stopwords were removed and remaining tokens were lemmatised using `WordNetLemmatizer`.
- Tokens shorter than 3 characters were discarded to reduce noise.
- The result was stored as Cleaned Review, alongside Review Length, Word Count, Character Count, and Average Word Length features.

3.4 Feature Engineering

- Sentiment was derived from Rating: Positive (rating ≥ 4), Neutral (rating = 3), Negative (rating ≤ 2).
- Age Group was created by binning Age into six bands: 18–25, 26–35, 36–45, 46–55, 56–65, and 66+.
- Has Positive Feedback was flagged when Positive Feedback Count > 0 .
- Long Review was flagged when Review Length exceeded the dataset median.

3.5 Data Preprocessing for Modelling

- The cleaned review text (Cleaned Review) was used as the sole model input; Recommended IND was used as the binary target.
- An 80% training / 20% testing split was applied with stratification on the target to preserve class balance (`random_state = 42`).

- Text was vectorised using TF-IDF (`max_features = 7000`, `min_df = 3`, `max_df = 0.85`, unigrams and bigrams).

3.6 Feature Selection

A Chi-Square (χ^2) feature selection step (`SelectKBest`, `k = 1200`) was applied on top of the TF-IDF matrix to retain only the text features most strongly associated with the Recommended IND target, improving model efficiency and reducing noise from rare or uninformative terms.

3.7 Model Selection

Four classification approaches were trained on the same TF-IDF $\rightarrow$ Chi-Square feature pipeline, each wrapped with class balancing where supported:

- Logistic Regression: Linear probabilistic baseline with `class_weight = "balanced"`.
- Multinomial Naive Bayes: Classic generative baseline for TF-IDF text classification.
- Linear SVM (`LinearSVC`): Margin-based linear classifier with `class_weight = "balanced"`.
- Random Forest Classifier: Ensemble tree model (150 estimators, `class_weight = "balanced_subsample"`) to capture non-linear interactions between text features.

3.8 Ethical Consideration

All analyses are based on voluntarily submitted, de-identified customer reviews. No personally identifiable customer information was used in model training, and the dataset is used strictly for aggregate customer-experience and text-analytics purposes.

4. Exploratory Data Analysis (EDA) & Visualizations

4.1 Data Quality and Target Distribution

- A missing-value heatmap confirmed that all null values were resolved after the cleaning stage.
- The target variable is imbalanced: 19,314 reviews ($\approx 82\%$) are Recommended (1) versus 4,172 ($\approx 18\%$) Not Recommended (0).

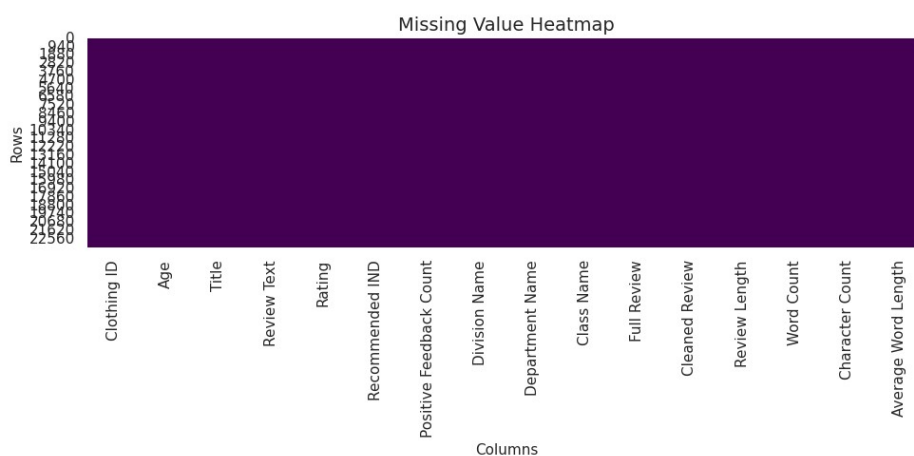


Figure 4.1a — Missing Value Heatmap (post-cleaning)

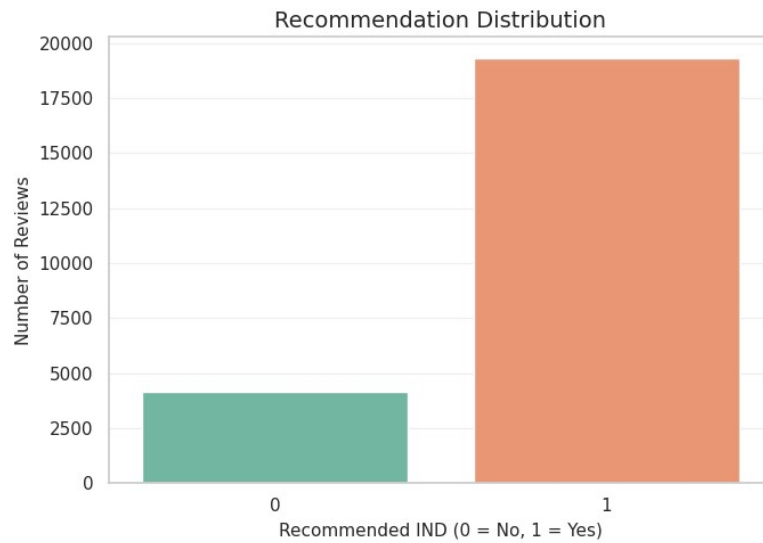


Figure 4.1b — Recommendation Distribution (0 = No, 1 = Yes)

4.2 Rating and Sentiment Distribution

- Ratings are heavily skewed toward 4 and 5 stars, consistent with the high overall recommendation rate.
- Derived sentiment shows 18,208 Positive, 2,871 Neutral, and 2,407 Negative reviews.

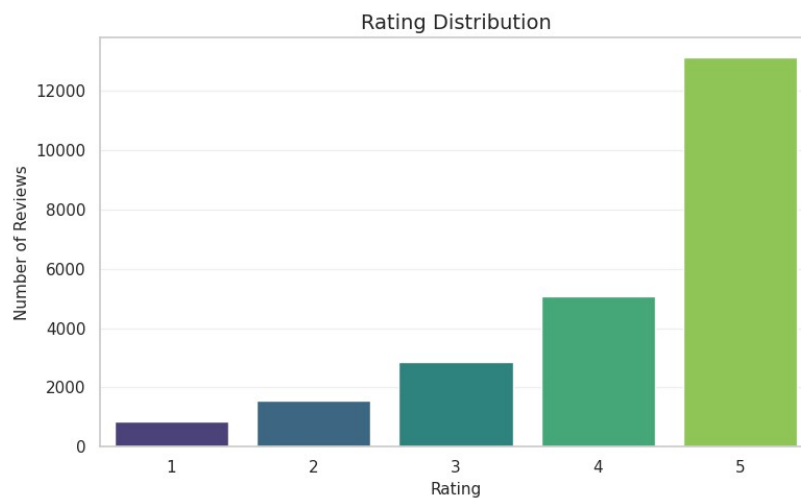


Figure 4.2a — Rating Distribution

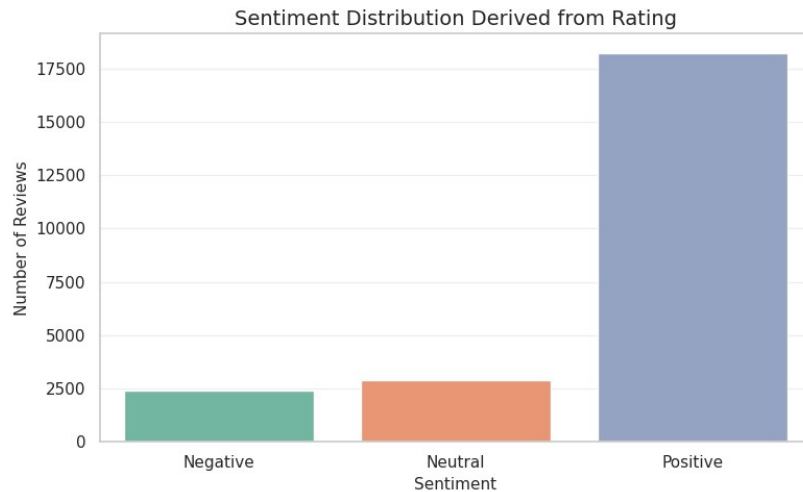


Figure 4.2b — Sentiment Distribution Derived from Rating

4.3 Customer Demographics

- Customer age is broadly distributed with a concentration between the mid-30s and mid-40s.

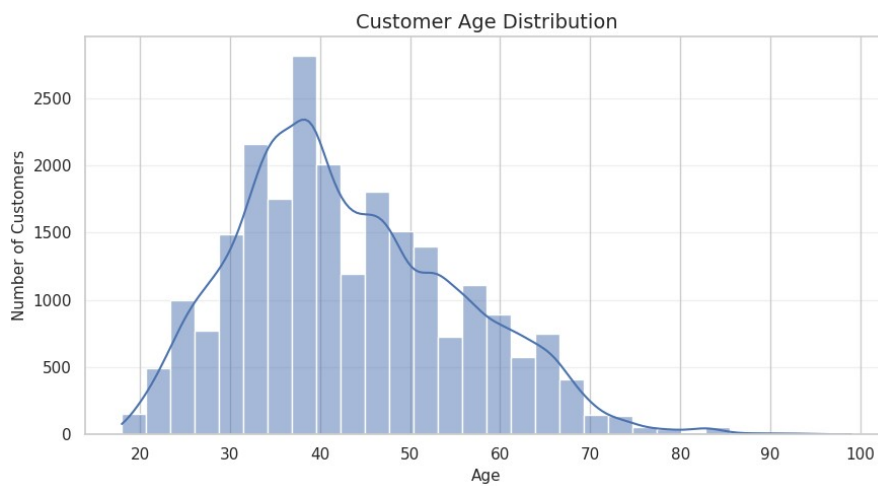


Figure 4.3 — Customer Age Distribution

4.4 Product Hierarchy Analysis

- Tops and Dresses dominate review volume across departments and divisions.
- Class-level distribution mirrors department trends, with a long tail of niche classes.

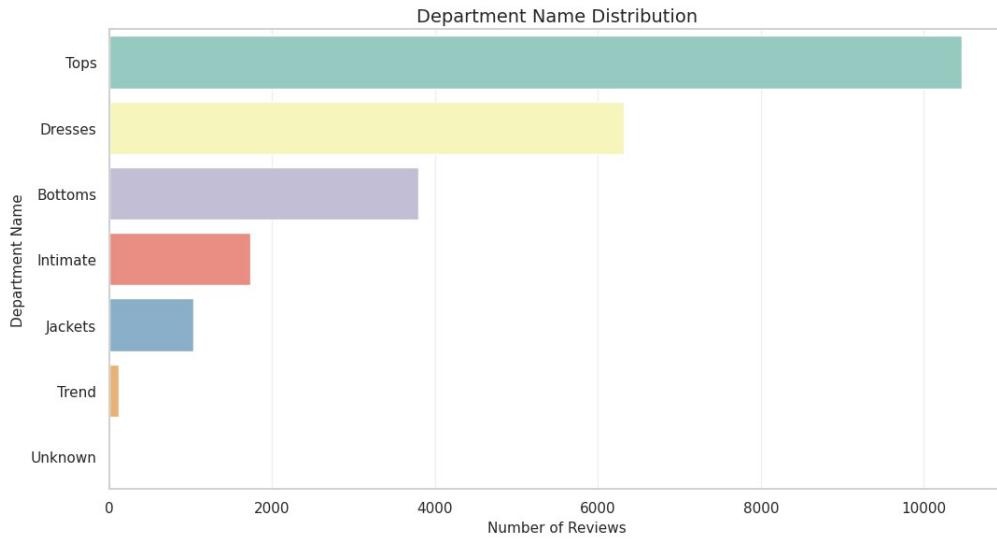


Figure 4.4a — Department Name Distribution

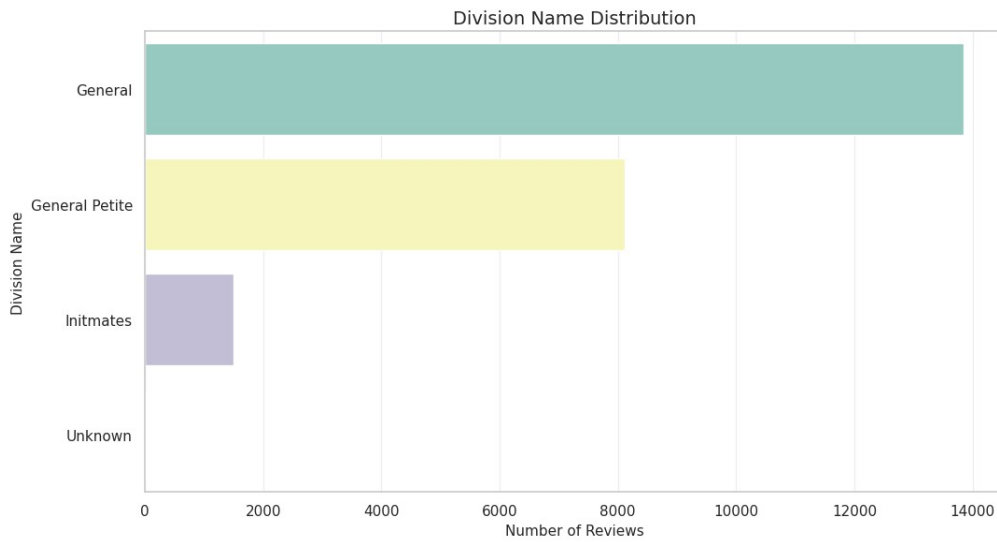


Figure 4.4b — Division Name Distribution

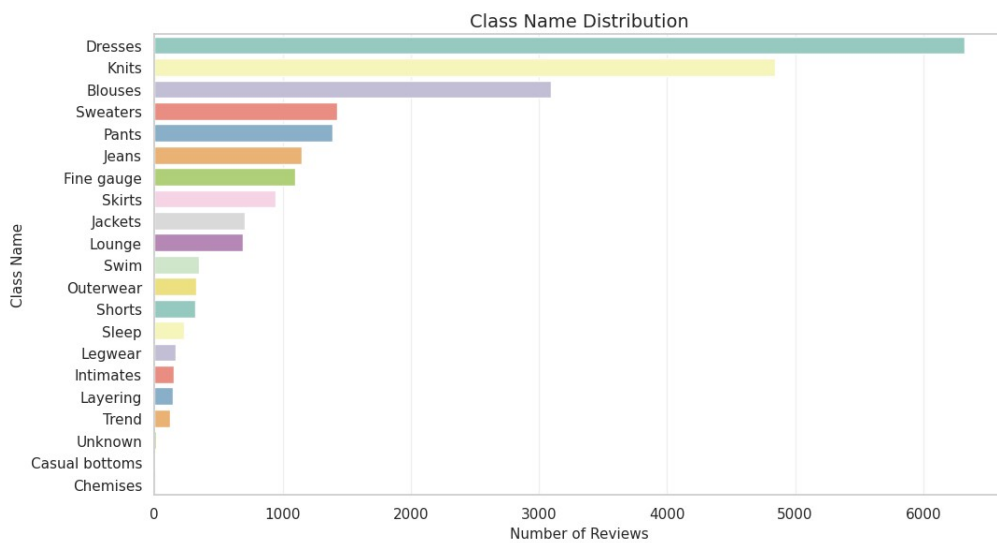


Figure 4.4c — Class Name Distribution

4.5 Rating Patterns by Product Hierarchy

- Average rating varies noticeably by department, with the Trend department showing the lowest average rating (3.82) and standard departments clustering higher.
- Division-level averages are more stable but still show measurable spread.

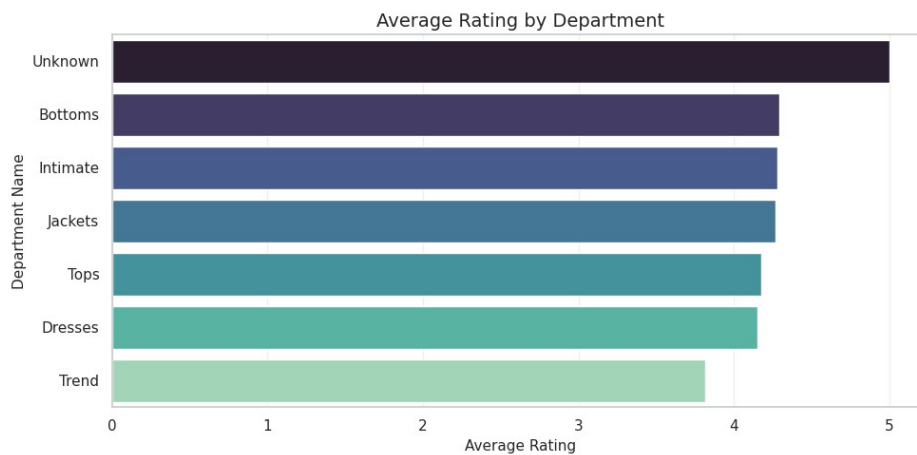


Figure 4.5a — Average Rating by Department

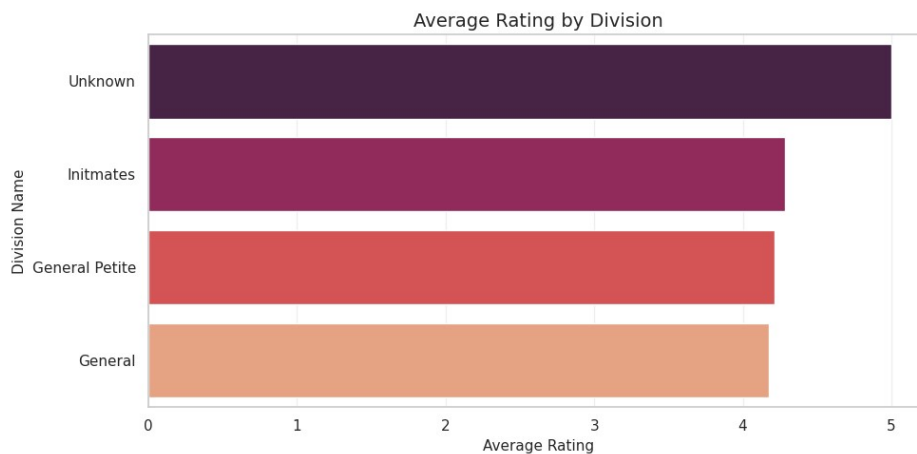


Figure 4.5b — Average Rating by Division

4.6 Feedback Engagement Analysis

- Positive Feedback Count is right-skewed — most reviews receive little to no helpful votes, with a long tail of highly-endorsed reviews.
- Feedback engagement rises modestly with more extreme ratings (both very low and very high), suggesting polarising reviews attract more attention.

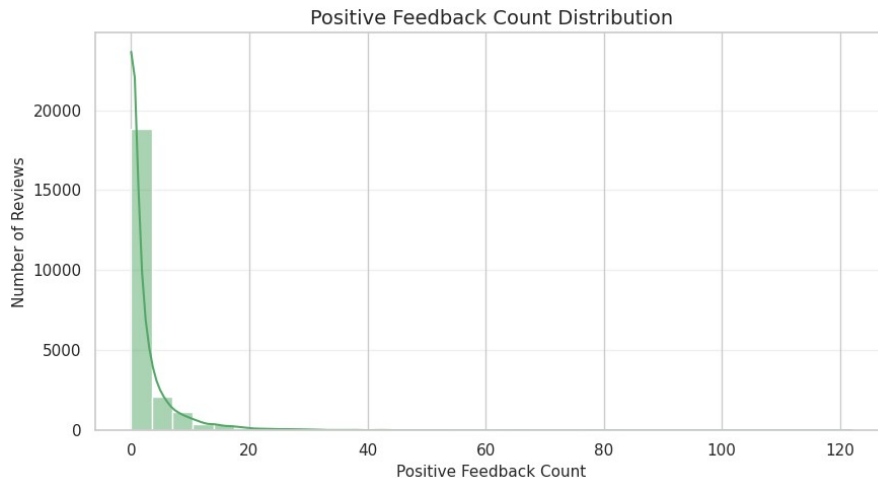


Figure 4.6a — Positive Feedback Count Distribution

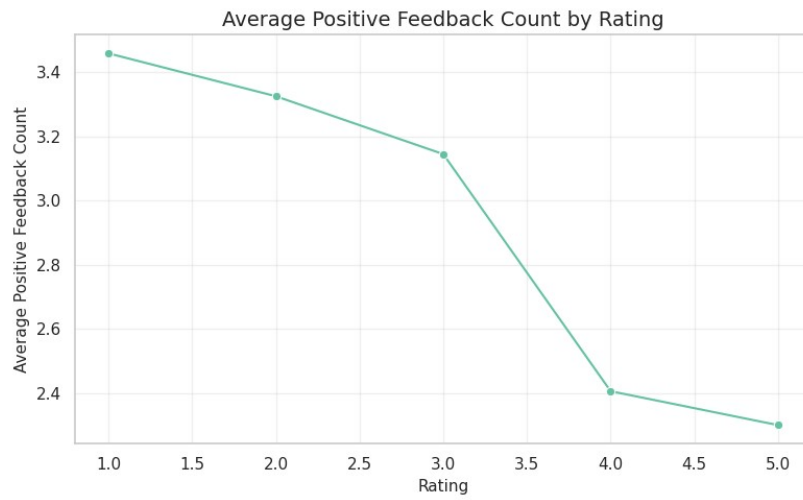


Figure 4.6b — Average Positive Feedback Count by Rating

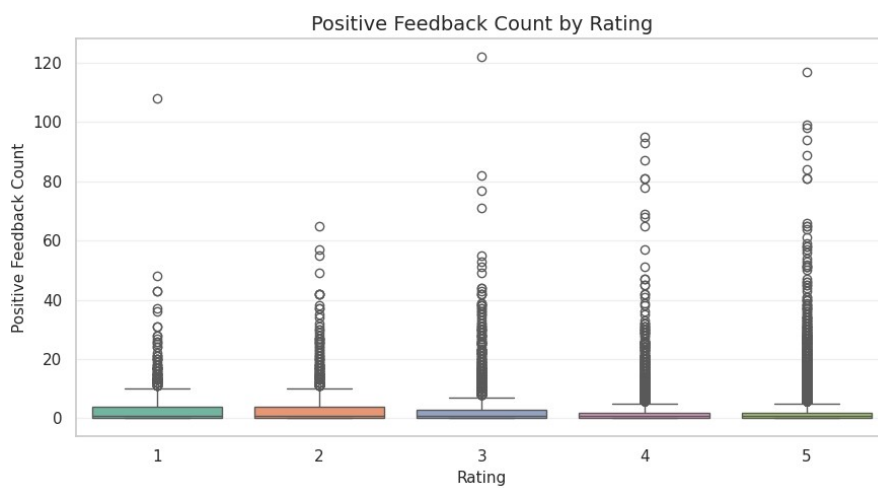


Figure 4.6c — Positive Feedback Count by Rating

4.7 Review Length Analysis

- Review length is right-skewed, with most reviews falling in a moderate length band and a tail of long, detailed reviews.
- Reviews associated with non-recommendation tend to run slightly longer — dissatisfied customers write more to explain their concerns.

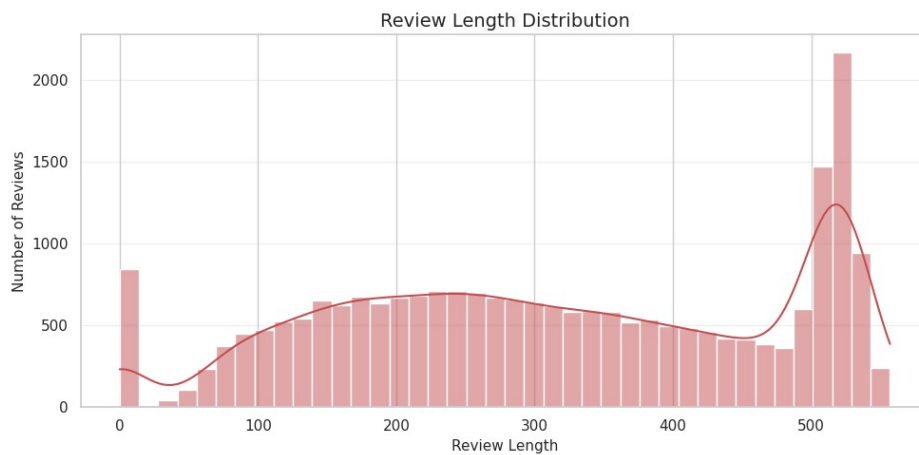


Figure 4.7a — Review Length Distribution

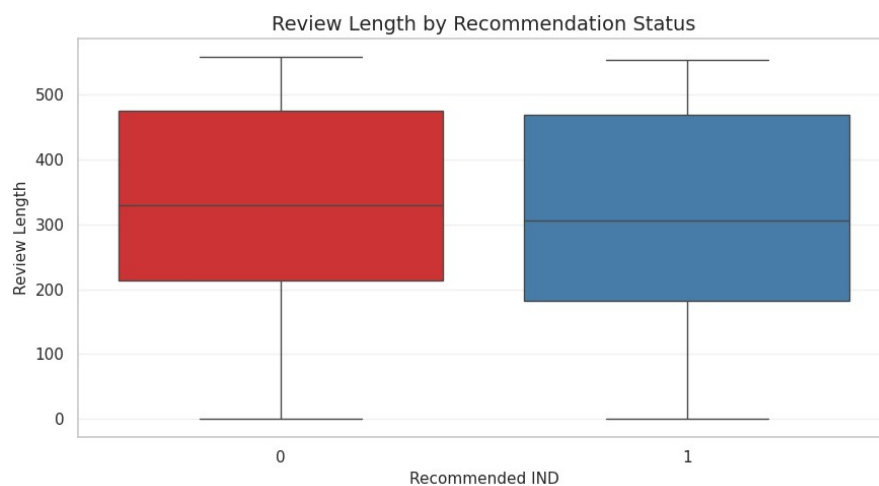


Figure 4.7b — Review Length by Recommendation Status

4.8 Correlation and Relationship Analysis

- Numeric correlation analysis shows the engineered text-length features (Review Length, Word Count, Character Count) are strongly inter-correlated, as expected.
- Rating shows a meaningful positive relationship with Recommended IND, confirming the two fields are well aligned.

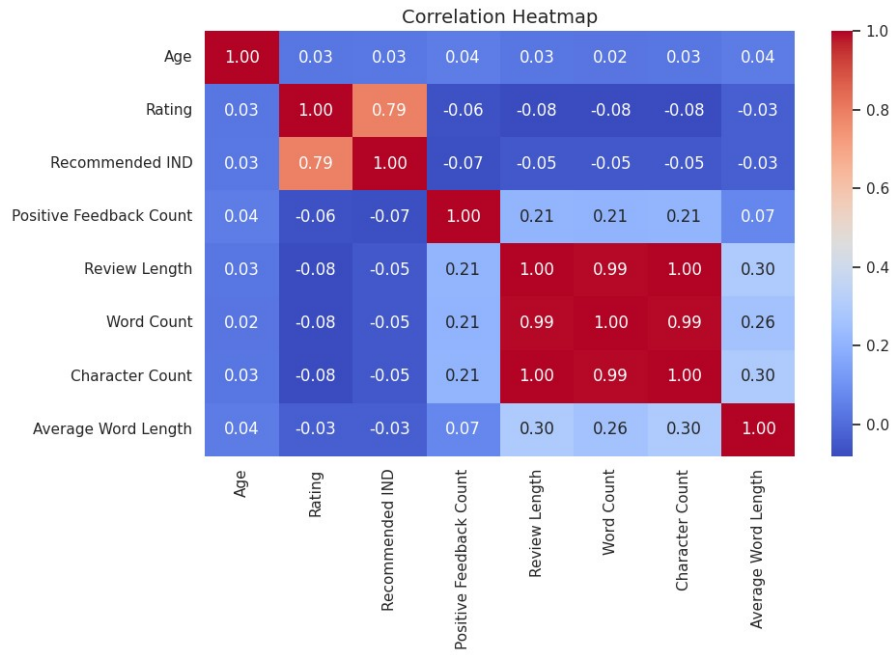


Figure 4.8a — Correlation Heatmap of Numeric Review Features

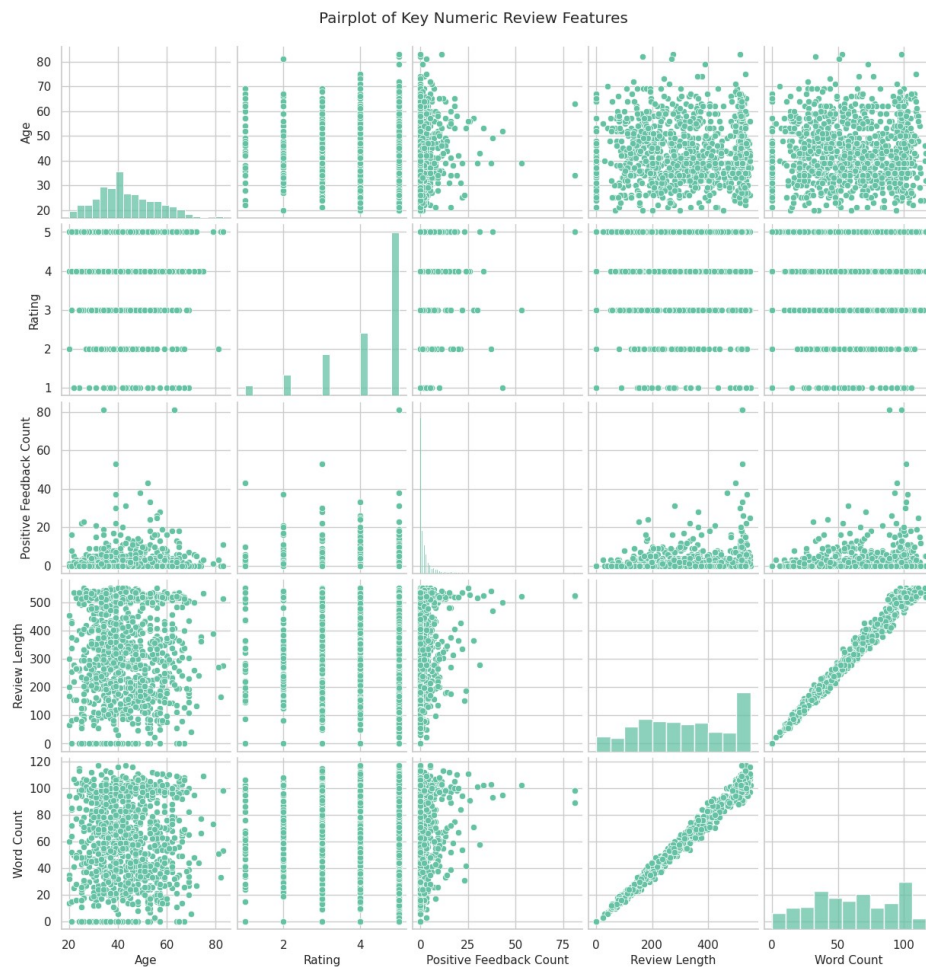


Figure 4.8b — Pairplot of Key Numeric Review Features

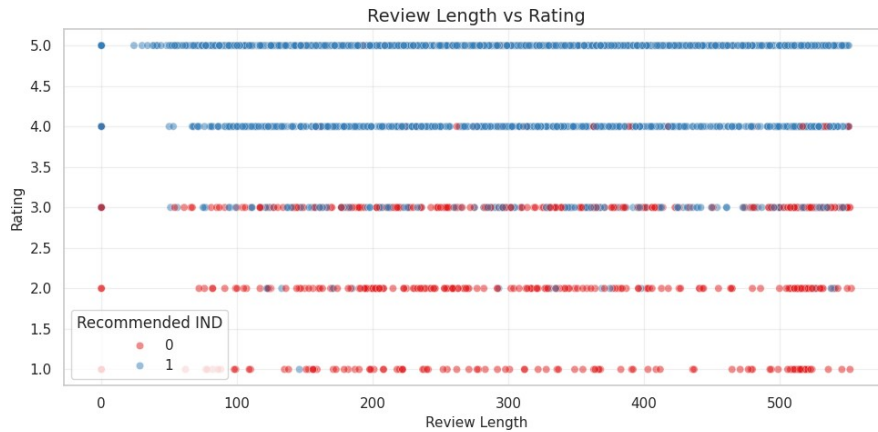


Figure 4.8c — Review Length vs Rating

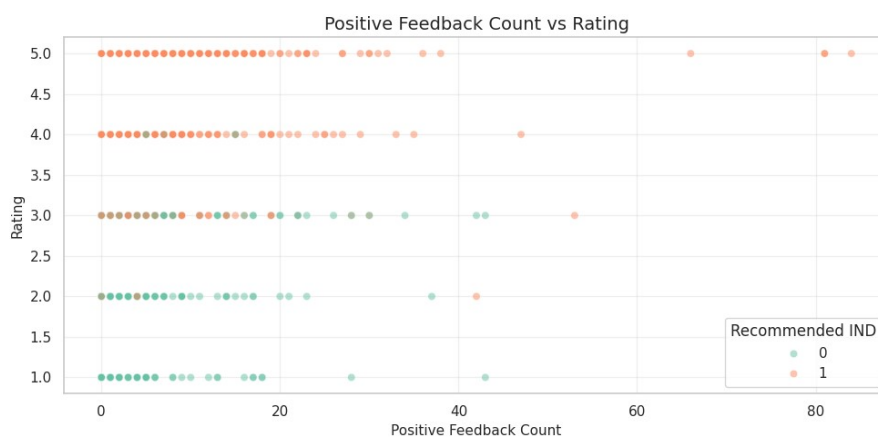


Figure 4.8d — Positive Feedback Count vs Rating

4.9 Text and Sentiment Insights

- The most frequent words across all reviews — dress, love, fit, size, top — reflect the fashion-retail domain and reveal that fit and sizing are dominant themes in customer feedback.
- A word cloud provides a quick visual summary of dominant review vocabulary.
- Sentiment composition varies by department: Trend has the highest negative-review share (18.5%), while Intimate, Jackets, and Bottoms are the most positively reviewed departments (all above 80% positive).

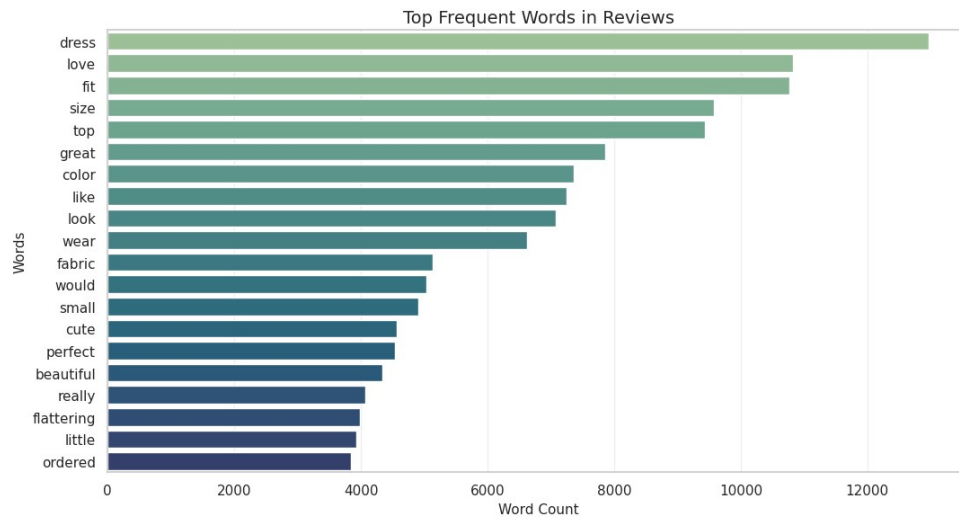


Figure 4.9a — Top Frequent Words in Reviews



Figure 4.9b — Word Cloud of Customer Reviews

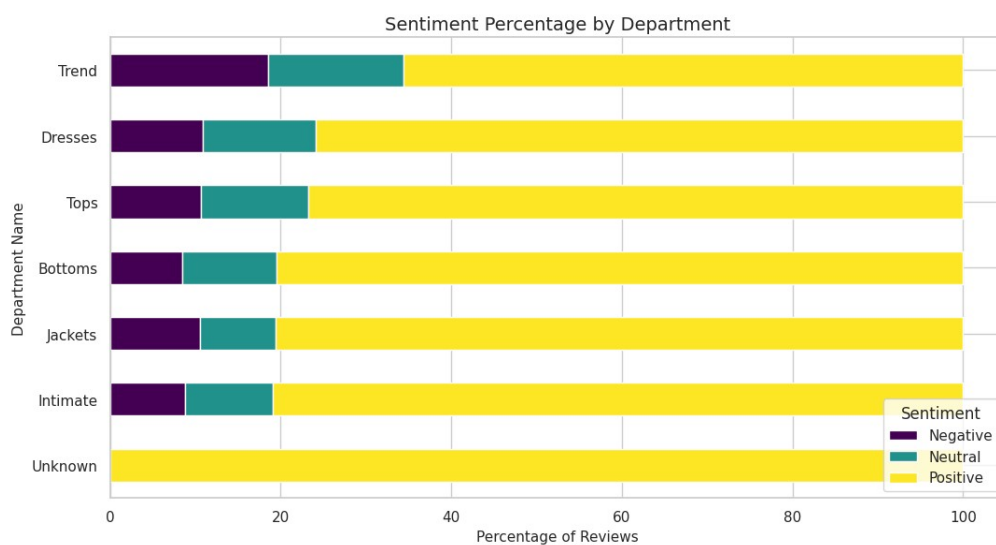


Figure 4.9c — Sentiment Percentage by Department

5. Feature Selection & Model Development

5.1 Chi-Square Text Feature Selection

A Chi-Square test was run on the TF-IDF matrix against the Recommended IND target to identify the text features (words and bigrams) most strongly associated with recommendation status. The top-ranked terms combine strong positive language (e.g. "love", "perfect", "great fit") and strong negative/fit-related language, confirming that fit and satisfaction language are the dominant discriminators.

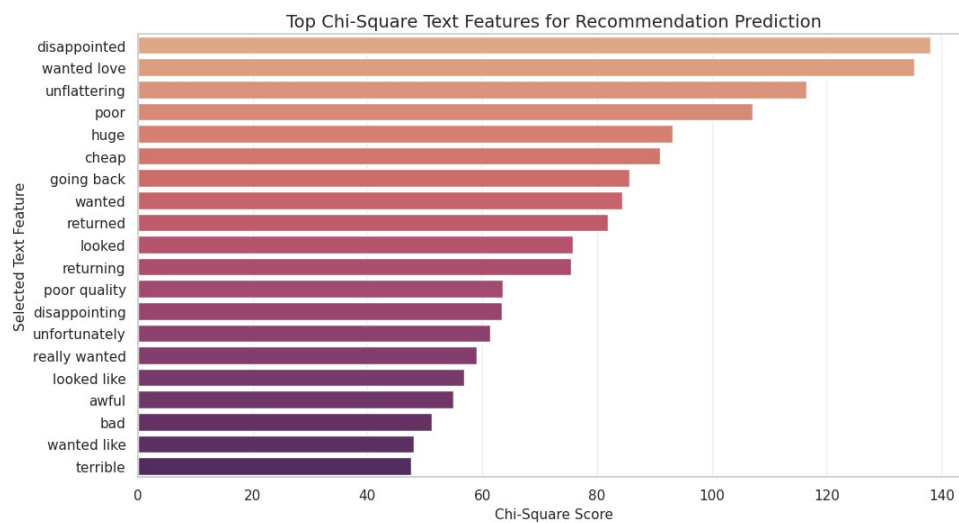


Figure 5.1 — Top Chi-Square Text Features for Recommendation Prediction

5.2 Model Training Setup

- All four models were trained on identical pipelines: TF-IDF (max_features = 7000, ngram_range = (1,2)) → Chi-Square selection (k = 1200) → classifier.
- Training set: 18,788 reviews · Testing set: 4,698 reviews (80/20 stratified split).
- Class imbalance (≈82% Recommended vs. ≈18% Not Recommended) was addressed via class_weight = "balanced" (or "balanced_subsample" for Random Forest); Multinomial Naive Bayes does not support class weighting.

5.3 Logistic Regression

Metric	Value	Interpretation
Accuracy	88.5%	Strong overall performance on imbalanced text data
Precision (Class 1 / Recommended)	0.97	Very few false positives among predicted recommendations
Recall (Class 0 / Not Recommended)	0.86	Good detection of dissatisfied customers
F1 Score	0.927	Well-balanced precision-recall trade-off
AUC Score	0.941	Excellent class separation, best AUC among all models

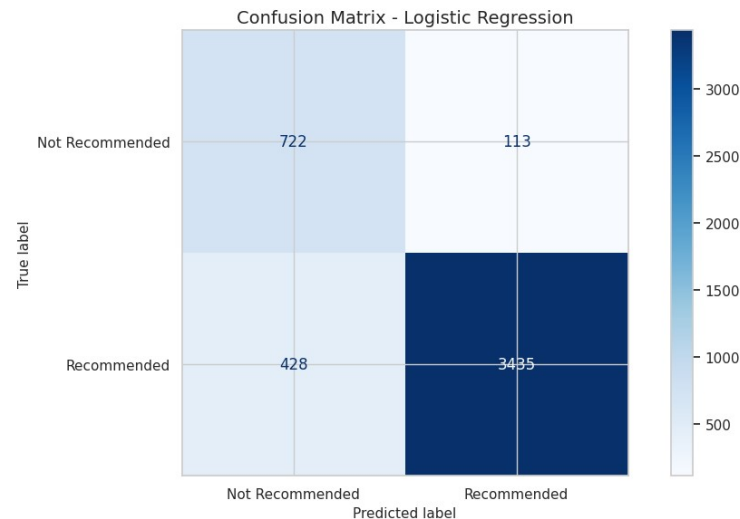


Figure 5.3 — Confusion Matrix: Logistic Regression

5.4 Multinomial Naive Bayes

Metric	Value	Interpretation
Accuracy	87.1%	Solid baseline given its simplicity
Precision (Class 0)	0.87	Reasonable precision on the minority class
Recall (Class 0)	0.33	Weakest recall — misses most non-recommended reviews
F1 Score	0.927	High overall F1, driven by majority-class performance
AUC Score	0.937	Good ranking ability despite weak minority recall

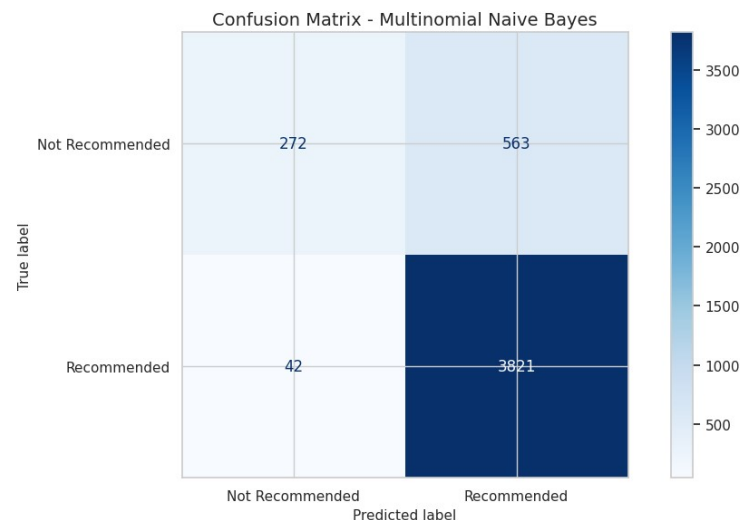


Figure 5.4 — Confusion Matrix: Multinomial Naive Bayes

5.5 Linear SVM

Metric	Value	Interpretation
Accuracy	88.4%	Comparable to Logistic Regression

Precision (Class 1)	0.96	Highly reliable positive predictions
Recall (Class 0)	0.84	Strong detection of dissatisfied customers
F1 Score	0.927	Well-balanced across both classes
AUC Score	0.941	Tied for the best class-separation performance

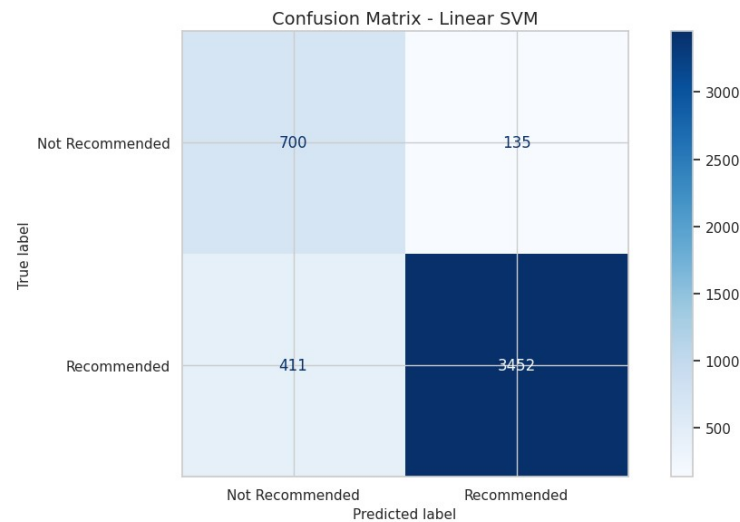


Figure 5.5 — Confusion Matrix: Linear SVM

5.6 Random Forest Classifier

Metric	Value	Interpretation
Accuracy	89.0%	Highest accuracy of all four models
Precision (Class 0)	0.71	Best minority-class precision among all models
Recall (Class 0)	0.64	Best minority-class recall among all models
F1 Score	0.934	Highest F1 Score — best overall balance
AUC Score	0.923	Strong, though slightly below the linear models

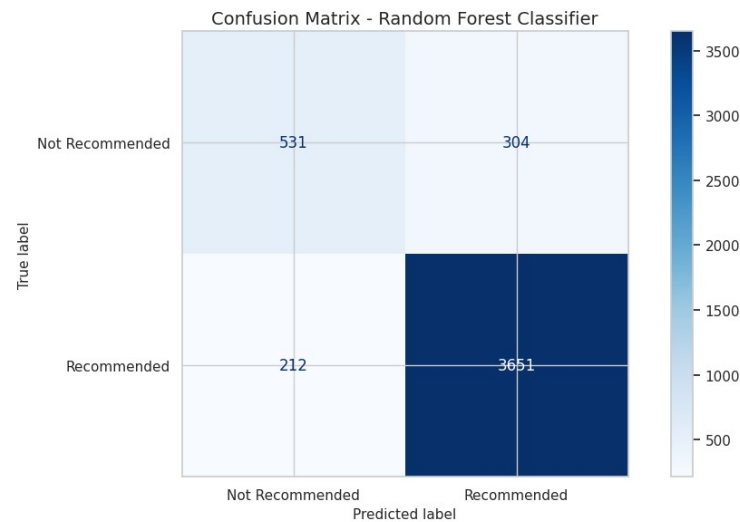


Figure 5.6 — Confusion Matrix: Random Forest Classifier

5.7 Model Comparison

The table below summarises all four models on the held-out test set (4,698 reviews):

Model	Accuracy	Precision	Recall	F1 Score	AUC
Random Forest Classifier	0.8902	0.9231	0.9451	0.9340	0.9226
Logistic Regression	0.8848	0.9682	0.8892	0.9270	0.9411
Linear SVM	0.8838	0.9624	0.8936	0.9267	0.9413
Multinomial Naive Bayes	0.8712	0.8716	0.9891	0.9266	0.9366

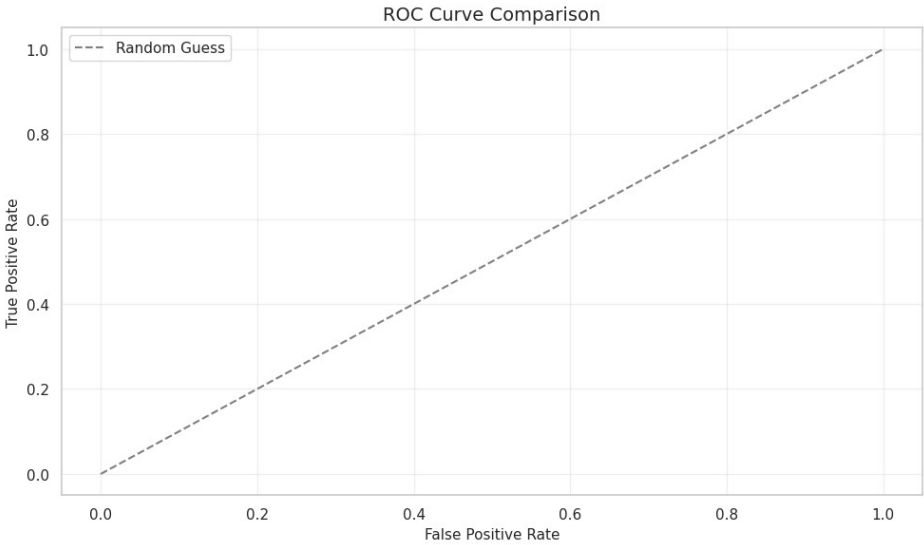


Figure 5.7a — ROC Curve Comparison Across Models

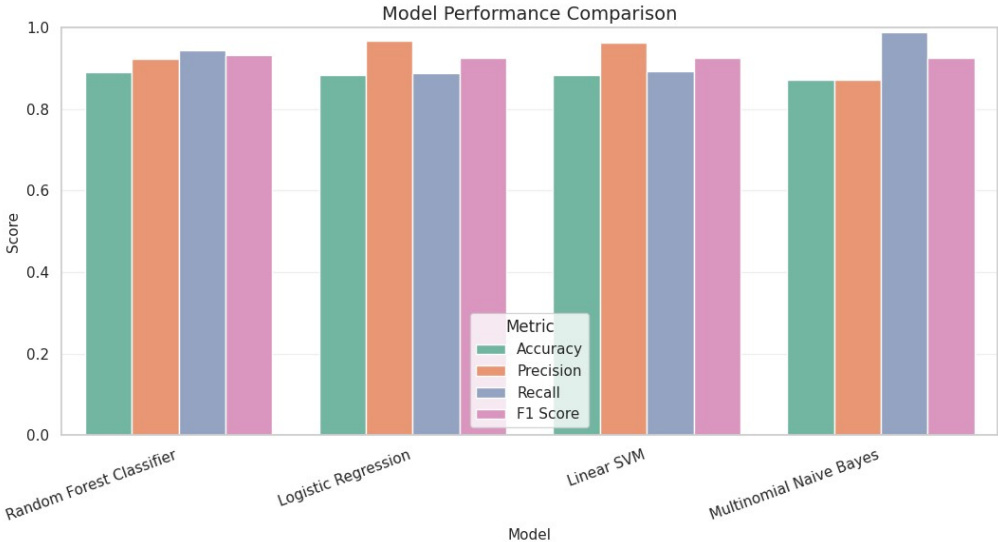


Figure 5.7b — Model Performance Comparison (Accuracy, Precision, Recall, F1)

Random Forest Classifier achieved the highest F1 Score (0.934) and was selected as the best-performing model, though Logistic Regression and Linear SVM offer marginally higher AUC and much higher precision on the

Recommended class — making them attractive simpler alternatives if interpretability or precision on positive predictions is prioritised.

6. Results & Conclusions

6.1 Logistic Regression

- Delivered the best AUC (0.941) tied with Linear SVM, with very high precision (0.97) on the Recommended class.
- Class balancing helped recall on the minority (Not Recommended) class reach 0.86 — the second-best among all models.
- Conclusion: A strong, interpretable baseline suitable when transparency in decision-making is important.

6.2 Multinomial Naive Bayes

- Fastest and simplest model, but recall on the Not Recommended class (0.33) is by far the weakest — it misses most dissatisfied customers.
- Conclusion: Useful only as a quick baseline; not recommended for production use given its poor minority-class detection.

6.3 Linear SVM

- Performance closely tracks Logistic Regression across all metrics, with a marginal edge in class-0 recall.
- Conclusion: A solid alternative to Logistic Regression with comparable reliability.

6.4 Random Forest Classifier

- Best-performing model overall: highest accuracy (89.0%) and highest F1 Score (0.934).
- Notably the strongest model at identifying actual Not Recommended reviews (recall 0.64, precision 0.71) — the metric that matters most for catching dissatisfied customers early.
- Conclusion: Recommended as the production model for automated recommendation prediction from review text.

6.5 Key Business Insights

- Fit and sizing language ("fit", "size", "small") dominate both praise and complaints — sizing consistency is the single largest lever for improving customer satisfaction.
- The Trend department has the lowest average rating (3.82) and the highest share of negative reviews (18.5%) — it warrants a focused quality or curation review.
- Intimate, Jackets, and Bottoms are the most positively reviewed departments (over 80% positive sentiment) and represent strong, stable product lines.
- Customers aged 26–35 give the lowest average ratings (4.14) while customers aged 56–65 give the highest (4.30) — younger shoppers may have higher expectations or different sizing needs.
- Non-recommended reviews tend to be longer, indicating dissatisfied customers write more explanatory detail — long, low-rated reviews are a useful signal for prioritising manual follow-up.

- With $\approx 82\%$ of reviews recommending the product, the dataset confirms generally strong customer satisfaction, but the Random Forest model's strength at isolating the $\approx 18\%$ who are not recommending makes it a practical tool for proactive customer-experience monitoring.

7. References

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— End of Report —